
THE STATE OF KAZAKH NLP: A DATA-DRIVEN SURVEY FROM GLOBAL BREAKTHROUGHS TO AN OPEN RESEARCH AGENDA

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ABSTRACT

Kazakh, an agglutinative Turkic language spoken by roughly 13 million people, has moved in a few years from near-absence in the NLP literature to a small but rapidly growing research field. Yet existing surveys treat Kazakh only as one of several Central Asian Turkic languages, leaving newcomers without a field-level map of what has been done in Kazakh specifically and where the open problems lie. We present the first survey dedicated to Kazakh, and the first built by a *reproducible*, citation-grounded pipeline rather than hand-curation: we automatically assemble a corpus of **222** Kazakh-focused papers (2013–2026) from the arXiv and Semantic Scholar APIs and analyse the resulting citation graph. Three findings organise the field. First, Kazakh NLP exhibits a **4–7 year adoption lag** behind global architectural breakthroughs (the Transformer appears in 2017 but the first large Kazakh LLMs only in 2024–2025). Second, the field rests on a handful of *dataset hubs*—the Kazakh Speech Corpus is cited 14 times within the corpus—and on four global pillars (the Transformer, BPE, BERT, and XLM-R). Third, the field is both young and lopsided: **47%** of all papers (105 of 222) appeared in 2024–2026, while across all years effort concentrates in speech (110 papers) and machine translation (105), leaving tokenization (21) and morphology (48) a thin, under-charted frontier. We close with an actionable research agenda of cited pain points and unverified hypotheses aimed at first-time contributors. All code, the corpus, and an interactive atlas are released to support reproduction and extension.

1 Introduction

Large language models have reorganised natural language processing around a small set of architectural and training breakthroughs, but the benefits of those breakthroughs reach the world’s languages unevenly. Kazakh—an agglutinative Turkic language with rich suffixal morphology and vowel harmony—is a representative case of a *moderately resourced* language: too central to a nation-state to be ignored, too far from the high-resource core to be well served by multilingual models trained predominantly on English.

In the span of roughly five years, Kazakh NLP has grown from scattered speech-recognition experiments into a field with dedicated corpora, named-entity and question-answering datasets, sentiment resources, and—most recently—several from-scratch and adapted large language models. This growth has been fast and largely uncoordinated, and no survey has yet stepped back to ask: *what, exactly, has been built, in what order, on whose shoulders, and what remains open?*

This paper answers that question with three contributions:

1. **A reproducible, data-driven survey method.** Rather than hand-pick papers, we assemble the corpus automatically from public APIs and release the scraping and analysis code, so the survey can be re-run and extended as the field grows (Section 2).

2. **A quantitative genealogy of the field.** Using the citation graph, we order the work as a citation-derived sequence of breakthroughs, surface the dataset and architectural hubs the field depends on, and measure the adoption lag behind global research (Sections 3–6).
3. **An open research agenda.** We compile cited pain points and unverified hypotheses scoped to be tractable for first-time contributors, turning a passive survey into a roadmap (Section 7).

To our knowledge this is the first survey dedicated specifically to Kazakh, and the first built by a reproducible, citation-grounded pipeline. The closest prior work surveys Kazakh as one of four Central Asian Turkic languages [24] or reviews Turkic NLP at large [25]; others treat Kazakh as one of many languages in a broader cross-lingual study—e.g. the 70-language morphological-alignment study of Arnett et al. [8]. None provides the field-level, single-language, citation-grounded genealogy we present.

2 Methodology

Corpus construction. We query the arXiv API and the Semantic Scholar Graph API (both accessed in June 2026) for Kazakh-related NLP/ML work. A paper is retained only if (i) its title or abstract mentions *Kazakh* or *Qazaq*, and (ii) it is relevant to NLP/ML (a keyword filter over tasks such as language modelling, tokenization, morphology, translation, and speech). Records are de-duplicated across sources by arXiv id, DOI, or normalised title. The resulting corpus contains **222** papers spanning 2013–2026; because of the API coverage limits noted below, all counts in this survey are best read as lower bounds on the true literature rather than exact censuses.

Citation graph. For each resolvable paper we retrieve its reference list via the Semantic Scholar batch endpoint and keep edges whose target is also in the corpus (*intra-corpus* edges) or is one of a small set of global breakthrough papers. This yields **312** edges (182 intra-corpus and 130 to global hubs); 205 of 222 papers were resolved and 104 form the connected core. We use in-degree (citations received within the corpus) as a measure of local influence and a topological view of the graph to order the field chronologically and logically (Section 4).

Task tagging. Each paper is assigned one or more of ten task categories (tokenization, morphology, language modelling, machine translation, speech, NER/IE, dataset/corpus, evaluation, classification, embeddings) by a keyword heuristic over title and abstract.

Reproducibility and scope. The scrapers, the corpus, the citation graph, and an interactive atlas are released as a public artifact.¹ We note two limitations of the method up front (expanded in Section 8): the Semantic Scholar API was intermittently rate-limited, so coverage of non-arXiv venues in some categories is incomplete; and category tags are heuristic rather than manually adjudicated.

3 Global Foundations

Kazakh NLP is built almost entirely on architectures developed elsewhere. The citation graph makes the dependence explicit: the most-cited external works, ranked by how many Kazakh papers reference them, are the Transformer [1] (23 citing papers), byte-pair encoding for NMT [3] (17), BERT [2] (16), XLM-R [4] (15), attention-based translation [5] (12), and GPT-3 [6] (10). Tokenization research enters the picture through Sennrich et al.’s subword units [3] and, later, fertility-based analyses of multilingual tokenizers [7]. These four “pillars”—a contextual architecture (Transformer/BERT), a multilingual encoder (XLM-R), a subword algorithm (BPE), and a scaling recipe (GPT-3)—account for the bulk of external influence on the field.

4 A Genealogy of Kazakh NLP

Ordering the citation graph topologically yields a reproducible sequence of breakthroughs that we group into five eras. This order is one of several valid topological sortings; we break ties by publication year, so the sequence below should be read as a structural narrative rather than a unique ranking.

Era I — Speech first (2015–2019). Kazakh NLP grew out of speech and translation rather than text understanding. The earliest hubs are bilingual Kazakh–Russian ASR systems and English–Kazakh NMT efforts that build directly on attention and BPE [5, 3, 1].

¹Code, corpus, citation graph, and interactive atlas: <https://github.com/altairzhambyl/kazakh-nlp-survey>.

Task category	Papers tagged
Dataset / corpus	146
Speech (ASR / TTS)	110
Machine translation	105
Evaluation / benchmark	76
NER / information extraction	55
Language modelling / LLM	50
Morphology / segmentation	48
Classification / sentiment	26
Embeddings	22
Tokenization	21

Table 1: Corpus by task category. Tags are multi-label, so a paper may be counted under several categories: the column reports the number of papers carrying each tag and sums to 659 tag occurrences across the 222 papers, not to 222. Tokenization and morphology—most relevant to Kazakh’s agglutinative structure—are the least studied.

Era II — The foundational corpus (2020). The single most influential resource is the crowdsourced Kazakh Speech Corpus (KSC) [10], cited 14 times within our corpus—more than any other Kazakh work. It converts speech recognition from a bespoke effort into a shared benchmark and seeds most of what follows.

Era III — The dataset-building years (2021–2022). A wave of open resources builds on KSC: KazNERD for named-entity recognition [11] (11 citations), the KazakhTTS [12] and KazakhTTS2 [13] text-to-speech corpora (7 and 8), and the industrial-scale KSC2 [14] (7). The field is, structurally, *dataset-driven*: progress is gated by corpora, and a handful of them carry most downstream work.

Era IV — LLMs and evaluation (2024). Attention turns to large models and to measuring them. “Do LLMs Speak Kazakh?” [17] provides the first systematic evaluation of seven models (6 citations); KazQAD [15] and KazSAnDRA [16] add open-domain QA and sentiment, and KazMMLU [18] and TUMLU [19] bring knowledge benchmarks. These works depend jointly on the earlier datasets and on global encoders [2, 4].

Era V — From-scratch and adapted LLMs (2025–2026). The frontier is Kazakh-specific language models: Sherkala [21], the from-scratch SozKZ family [22], the tokenizer-free KazByte adapter [23], and the state-backed Til-Core.² The adoption lag is now quantifiable: the Transformer (2017) is cited throughout, but the first large Kazakh LLMs appear only in 2024–2025—a 4–7 year delay between a global breakthrough and its Kazakh realisation.

5 The Landscape by Task

Table 1 shows the distribution of the corpus across task categories. Because tags are multi-label, the figures are tag counts rather than a partition of the 222 papers. The shape is informative: Kazakh NLP is, historically, a *speech-and-translation* field. Dataset/corpus papers (146) dominate because resource construction is the binding constraint; speech (110) and machine translation (105) follow. Crucially, the two categories most relevant to the language’s defining property—its agglutinative morphology—are the thinnest: morphology (48) and tokenization (21). This is the central empirical observation of the survey: the field has invested heavily where data could be collected, and least where the language is hardest. The corpus is also sharply concentrated in time: 105 of the 222 papers (47%) carry a 2024–2026 date, against 18 in 2020 and 3 at the earliest point in 2013, confirming that Kazakh NLP is overwhelmingly a development of the last three years.

6 Models and Tokenizers

Table 2 compares the flagship Kazakh-facing models. Two patterns stand out. First, the field is converging on *Kazakh-specific tokenization*: vocabularies grow from 32k (KazRoBERTa) to 256k (Til-Core), and approaches diverge from extended BPE (Sherkala) to from-scratch BPE (SozKZ) to abandoning the tokenizer entirely (KazByte [23]). Second, claims are running ahead of evidence: the state-backed Til-Core advertises a morpheme-constrained tokenizer (“support

²Til-Core-0.5B model card: <https://huggingface.co/TilQazyna/Til-Core-0.5B>. At the time of writing it has no accompanying paper or published benchmark results.

Model	Year	Vocab	Morphology claim	Benchmarks
Til-Core-0.5B	2025	256k (morphBPE)	yes (segmenter unreleased)	none
Sherkala-Chat-8B	2025	159k	no (fertility-driven)	yes
SozKZ (50M–600M)	2026	50k (from scratch)	no	partial
KazByte	2026	byte-level (none)	n/a	yes
KazLLM (8B/70B)	2024	128k (Llama-3.1)	no	yes
KazRoBERTa	2022	32k	no	partial

Table 2: Flagship Kazakh-facing models. The “morphology claim” column reports authors’ assertions, not verified results. Entries without an accompanying paper (Til-Core, KazLLM, KazRoBERTa) are characterised from their public model cards rather than peer-reviewed work.

for Kazakh morphology”) but has, at the time of writing, *no published benchmark results*. Whether morphology-aware tokenization actually helps Kazakh remains an open question—and a contested one, since prior work on agglutinative languages finds morphological segmentation does not reliably beat BPE downstream [9].

7 An Open Research Agenda

We distil the survey into concrete entry points for new contributors, separating *pain points* (capabilities or resources the field lacks) from *unverified hypotheses* (claims that are asserted or assumed but not rigorously tested).

7.1 Pain points

Representative, tractable gaps include: the high *tokenization tax* on Kazakh, which motivates KazByte [23] and SozKZ [22] but is not measured downstream; the absence of a standardised *morpheme-boundary benchmark*; weak support for Kazakh–Russian *code-switching*, despite its prevalence in real usage; a *fragmented evaluation landscape* (KazMMLU, TUMLU, KazQAD, Qorgau use different protocols [18, 19, 15, 20]); near-absent ASR for spontaneous and children’s speech; and the failure of multimodal models on Kazakh Arabic/Latin-script OCR. Further tractable gaps include sentiment analysis beyond product reviews, intrinsic embedding benchmarks for Kazakh, and ASR punctuation restoration. The released agenda lists each with its motivating citation.

7.2 Unverified hypotheses

The field carries several load-bearing assumptions that a student could test: (1) morphology-aware segmentation improves downstream Kazakh tasks over BPE—*contested* by [9]; (2) byte-level tokenization beats BPE for agglutinative Kazakh—*unverified*, with KazByte’s authors noting validation is ongoing [23]; (3) cross-lingual transfer from Turkish beats transfer from Russian—*assumed* but never ablated; (4) larger tokenizer vocabularies meaningfully improve downstream quality—*unverified*, with no controlled study; (5) a small from-scratch Kazakh model beats a large multilingual one at equal inference budget—*unverified*. Each is stated with a concrete experiment in the released agenda.

8 Limitations

This survey is bounded by its method. Semantic Scholar rate-limiting left non-arXiv coverage incomplete in several categories, so counts are lower bounds rather than exact censuses. Task tags are heuristic and were not manually adjudicated. The citation graph captures only references resolvable through Semantic Scholar, so the connected core (104 papers) under-counts true dependencies. Finally, this survey synthesises the field from titles, abstracts, and citation structure rather than a close reading of every paper; the per-paper claims, numbers, and attributions in Sections 4–6 should therefore be read as a structural map of the field rather than a verified account of each individual result.

9 Conclusion

Kazakh NLP is young, fast-growing, and structurally lopsided: rich in speech and translation resources, thin where the language is morphologically hardest, and dependent on a small set of datasets and global architectures. By mapping the field reproducibly and ending with a concrete agenda, we hope to lower the barrier for the next wave of contributors—especially those making a first contribution—and to make the morphology and tokenization frontier, where the open questions are sharpest, easier to enter.

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